

1 ASIO Full Analysis Report

This report summarizes the analysis of dataset 20250919.0x0003. The trial spans 119.99 s of continuous data acquisition across all six channels. The remainder of this section reports both raw and detrended + EMI-filtered statistics. Units are femtoamps for mean/RMS/std; skew and kurtosis are dimensionless.

1.1 Detrended + EMI-filtered Statistics

Table 1: Detrended + EMI-filtered Current Statistics

Channel	Mean (fA)	RMS (fA)	Std Dev (fA)	Skew	Kurtosis
SXR1	1.0084×10^{-14}	2.6064×10^2	2.6064×10^2	-1.0610×10^0	2.2947×10^1
SXR2	3.5558×10^{-15}	5.2788×10^1	5.2788×10^1	1.3642×10^{-1}	3.8992×10^{-2}
SXR3	-2.8272×10^{-15}	5.6206×10^1	5.6206×10^1	3.4914×10^{-2}	-2.5522×10^{-2}
SXR4	1.2605×10^{-17}	5.2999×10^1	5.2999×10^1	-3.0852×10^{-2}	-7.2751×10^{-2}
HXR	6.8704×10^{-15}	7.0287×10^2	7.0287×10^2	6.3366×10^{-2}	2.6221×10^{-2}
EUV	1.9203×10^{-14}	8.2739×10^2	8.2739×10^2	-1.6752×10^{-2}	-1.2738×10^{-1}

1.2 Raw Statistics

Table 2: Raw Current Statistics

Channel	Mean (fA)	RMS (fA)	Std Dev (fA)	Skew	Kurtosis
SXR1	6.2439×10^5	6.2439×10^5	5.3957×10^2	-4.8794×10^{-1}	2.7318×10^0
SXR2	5.4093×10^4	5.4093×10^4	5.6982×10^1	1.8356×10^{-1}	1.2220×10^{-1}
SXR3	5.4307×10^4	5.4307×10^4	5.9627×10^1	2.8992×10^{-2}	-1.5393×10^{-2}
SXR4	5.3596×10^4	5.3596×10^4	5.6073×10^1	-6.2249×10^{-2}	-9.9629×10^{-2}
HXR	6.8387×10^5	6.8387×10^5	7.4840×10^2	1.0586×10^{-1}	1.0781×10^{-2}
EUV	7.7276×10^5	7.7276×10^5	1.0046×10^3	1.3260×10^{-1}	-3.4390×10^{-1}

1.3 Trial Duration

Quantity	Value
Total acquisition time	119.990 s

1.4 Raw Voltage Time Series

ASIO Raw Voltage Values

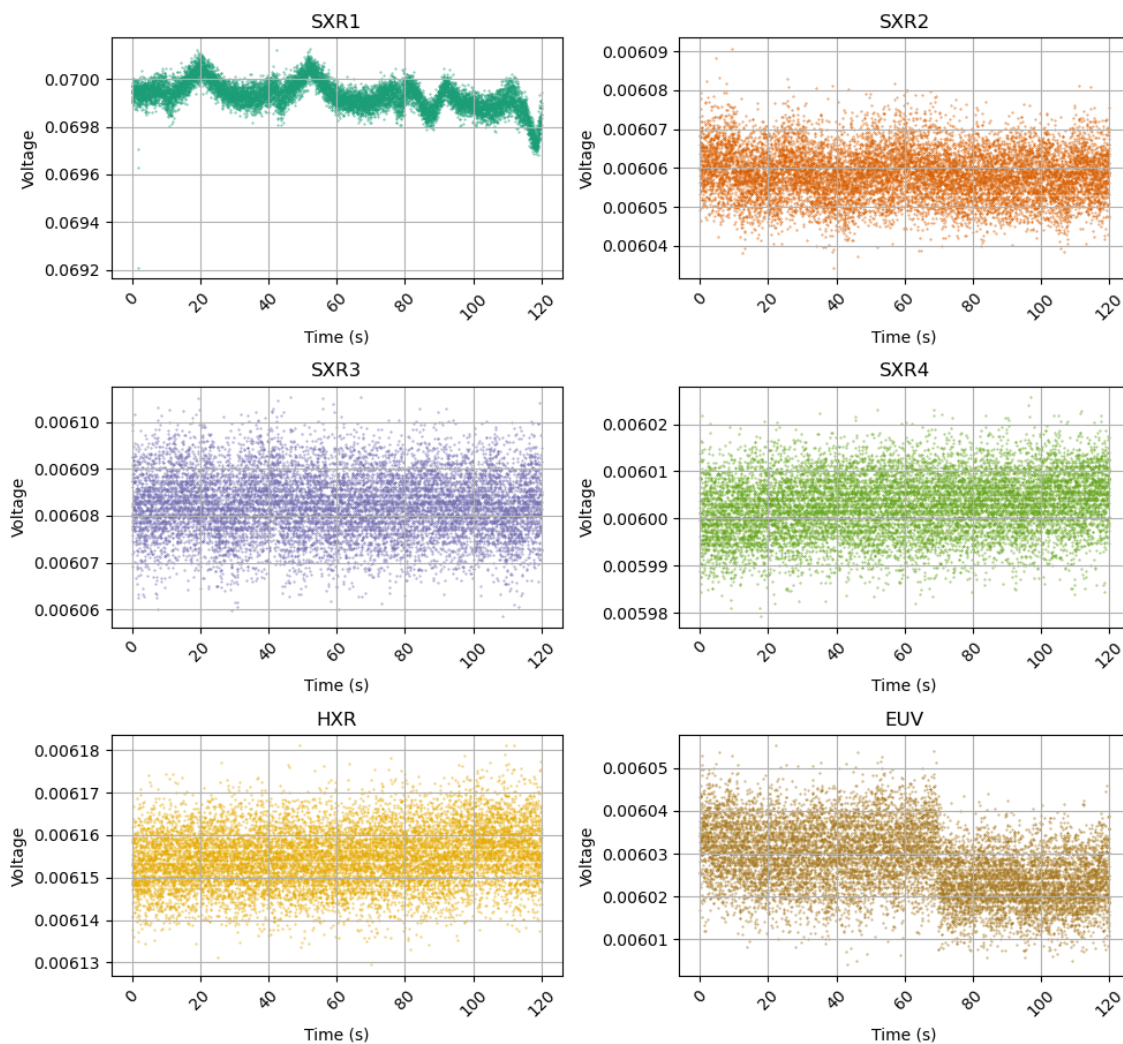


Figure 1: Raw voltage vs. time (s) for all six ASIO channels.

1.5 FFTs of Raw Voltages

FFTs

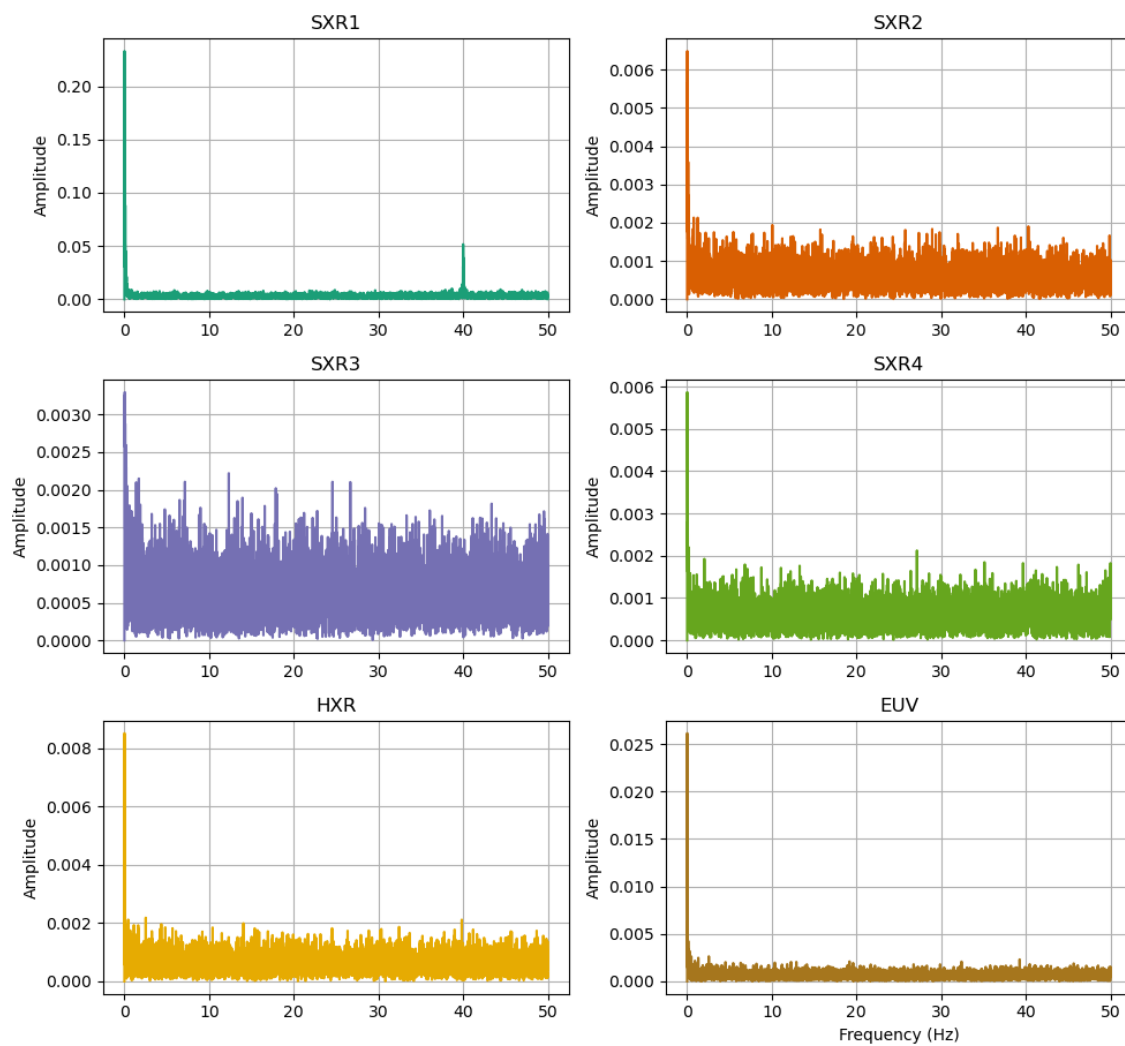


Figure 2: FFTs of raw voltages (mean centered).

1.6 Detrended + EMI-filtered Time Series

EMI-Filtered and Detrended Voltage Values

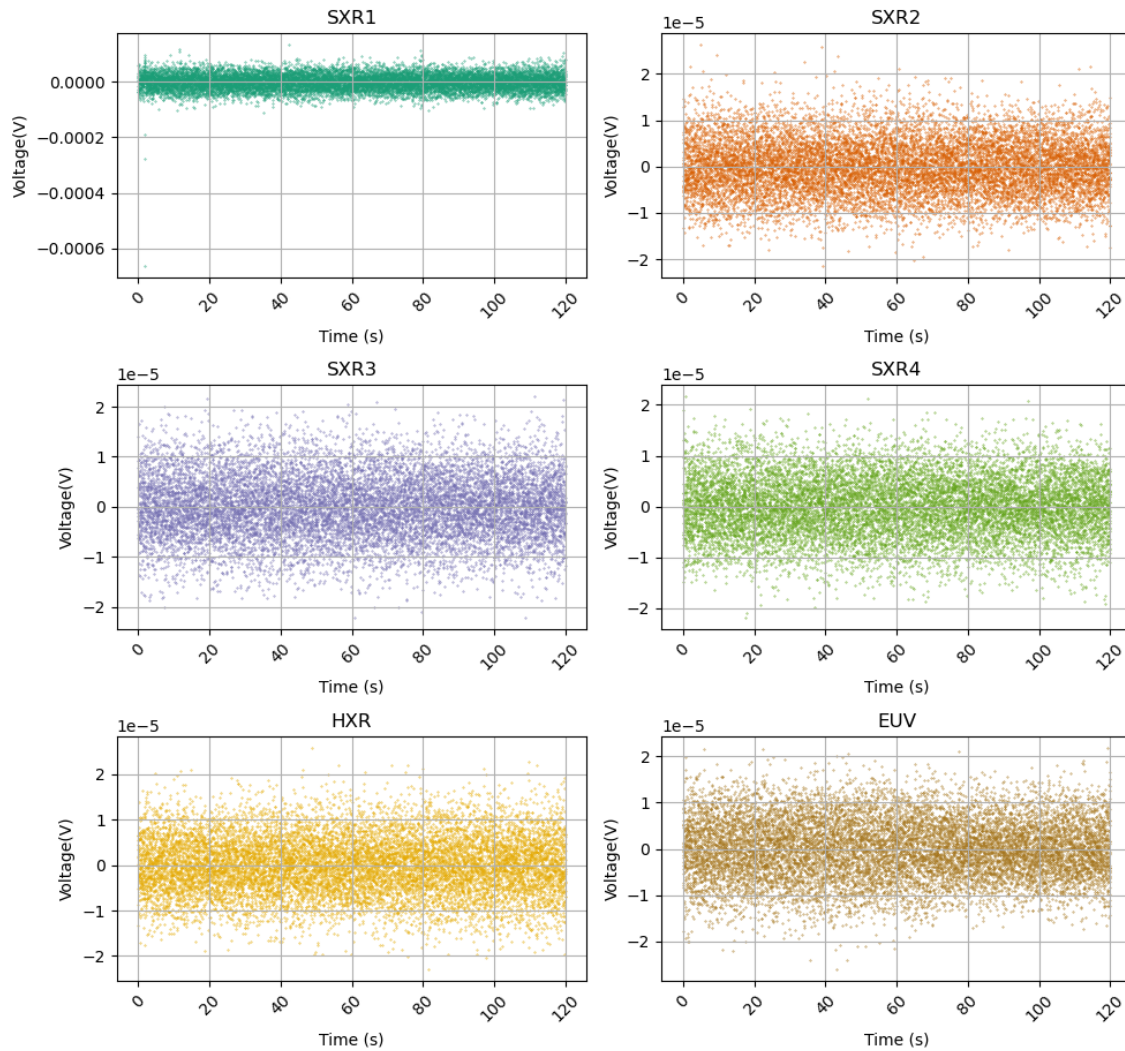


Figure 3: Voltage vs. time (s) after Savitzky–Golay detrending and 40 Hz EMI removal.

1.7 Noise Histograms

ASIO Noise Histograms w/EMI-filtering and Detrending

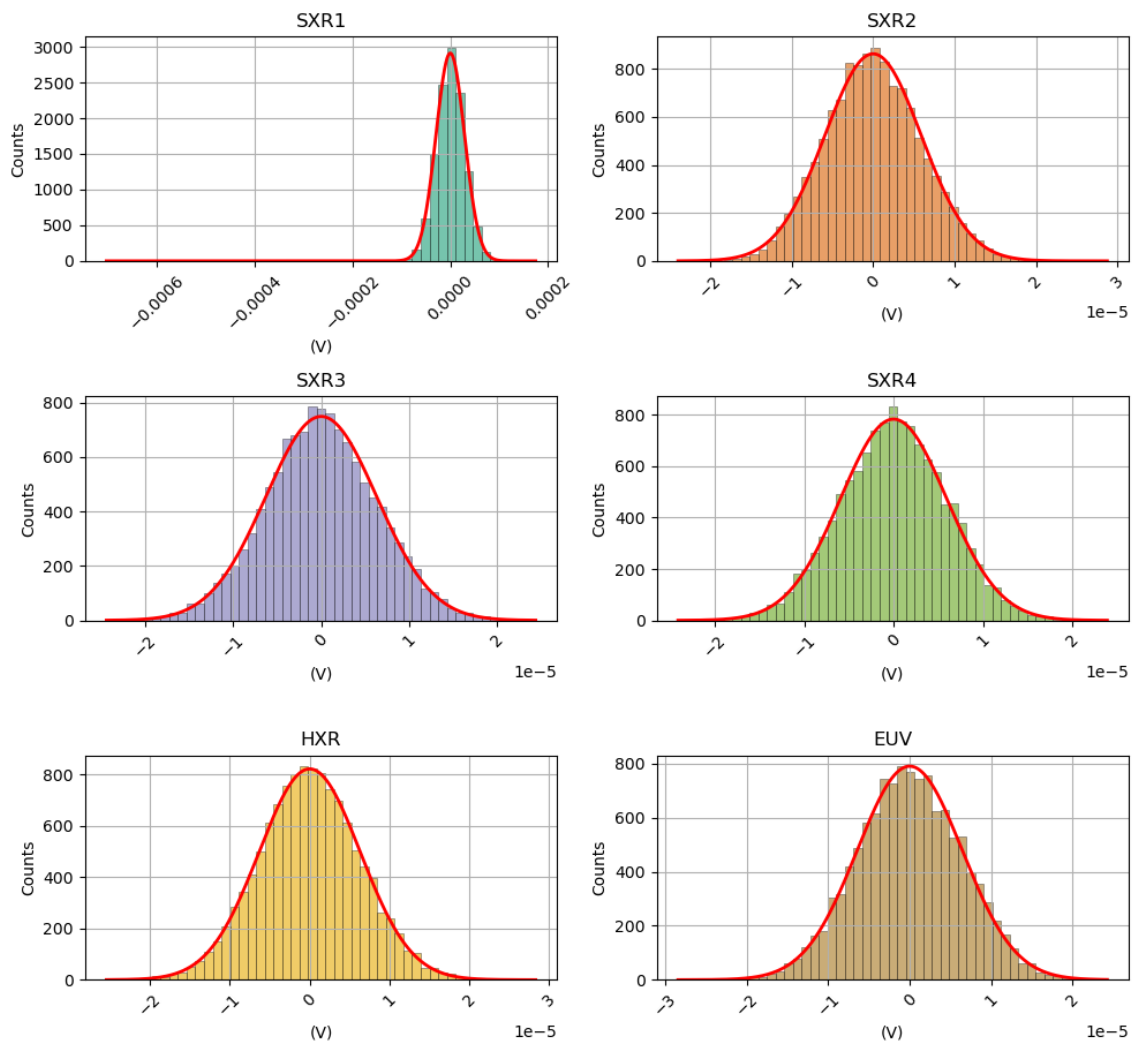


Figure 4: Noise histograms after detrending and 40 Hz EMI removal. Red curve is a Gaussian fit.

1.8 Fe-55 Expectation Values

Fe-55 expectation values: not yet implemented.

Testing Notes

selection: whole dataset (t=0.0s–120.0s)